BPM2_NB_ESOFT QUIZ 4_S2026

# of Questions: 30

Question #: 1

A 34-year-old woman comes to the physician because of a 1-day history of blurry vision to her right eye, and a 6-month history of mild, intermittent pain in the same eye. Physical examination shows a 0.3-mm ulceration to the right cornea. This region of the eye is normally characterized by the presence of cells that produce the collagen fibrils found in the corneal stroma. Which of the following best identifies these cells?

A. Keratocyte
B. Melanocyte
C. Keratinocytes
D. Endothelial
E. Goblet

Item Description: HCB.1334 - Structure of cornea - Keratocytes

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Question #: 2

A 34-year-old man is brought to the emergency department by the paramedics becuase he was unconscious for a few minutesafter a motor vehicle accident, about an hour ago. Physicial examination shows left eye is having trouble moving laterally (abduction) from a fully adducted horizontal position. Damage to which cranial nerve will most likely cause this condition?

A. Left cranial nerve VI
B. Right cranial nerve VI
C. Right cranial nerve III
D. Left cranial nerve III
E. Left cranial nerve VII
F. Right cranial nerve VII

Item Description: Cranial nerve VI

Question #: 3

A 6-year-old girl is brought to the physician because of a 2-week history of inability to see the blackboard in school. She is at the 95% percentile for height and is slender with long arms and fingers. Physical examination shows kyphoscoliosis. Slit lamp examination of the eyes shows inward and downward dislocation of the lens. Laboratory studies show early signs of atherosclerosis. She is prescribed pyridoxine supplements. Which of the following processes is most likely affected in this patient?

A. Catabolism of the branched chain amino acids
B. Absorption of tryptophan in the intestine
C. Conversion of homocysteine to cysteine
D. Synthesis of heme from glycine and succinyl CoA
E. Hydrolysis of TAG in Chylomicrons and VLDL
F. Beta oxidation of medium chain fatty acids

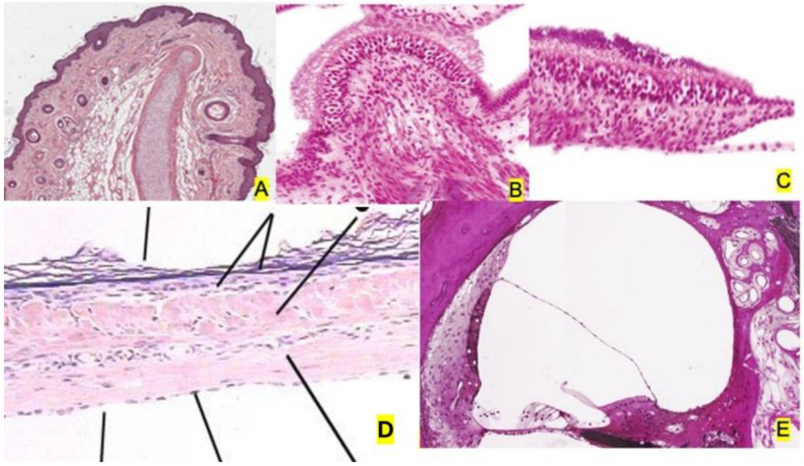
Item Description: Homocystinuria - quiz

Question #: 4

A 5-year-old girl is brought to the physician by her mother because of a 1-week history of fever, cough, and discharge from the right ear. Her temperature is 40°C (104°F). Physical examination shows a febrile child with sunken eyes, poor skin tugor, and yellow, viscous discharge from the right ear. Otoscopy of the right ear shows a perforation of the tympanic membrane, with a purulent discharge. Which of the following microanatomical images best corresponds with the perforated structure?

A. A
B. B
C. C
D. D
E. E

Attachment:



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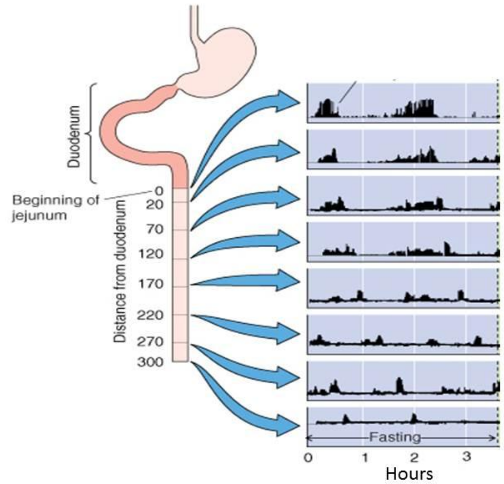
Item Description: HCB.11362 - Tympanic membrane

Question #: 5

A 22-year-old woman is brought to the emergency department by her mother because of a 2-hour history of altered mental state. The mother says that her daughter has not eaten any food for the past 2 days. Her blood pressure is 104/50 mm Hg and pulse is 112/min. Physical examination shows muscle weakness, dry mouth, sunken eyes, and reduced concentration. Abdominal examination shows a soft abdomen with intermittent bowel sounds. These bowel sounds are produced by sequential contractions of the gastrointestinal tract that originate from the proximal jejunum and migrate towards the cecum as shown in the attached figure. Which of the following hormones is most likely responsible for these contractions in this patient?

A. Cholecystokinin
B. Acetylcholine
C. Secretin
D. Vasoactive intestinal peptide
E. Motilin

Attachment:



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Item Description: PHYS.1113. MMC Hormone responsible

Question #: 6

A 56-year-old man comes to the physician because of a 4-month history of difficulty using his right hand. He says that a heavy piece of metal in his workshop had fallen on his arm, and he received a diagnosis of a crush injury to a nerve. Which of the following outcomes is most likely following successful reinnervation of the previously dennervated muscles?

A. Hypertonia
B. Hypotonia
C. Normotonia
D. Dystonia
E. Atonia

Item Description: SOM.MK.II.BPM2.4.NB.2.NSCI.0177 Denervation and reinnervation in lower motor neuron syndrome

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Question #: 7

A 66-year-old man comes to the physician because of a 6-day history of numbness in his left arm. Neurologic examination shows reduced somatosensation in the left arm. Strength and reflexes are normal. Which of the following gyri is most likely injured?

A. Inferior frontal
B. Inferior temporal
C. Cuneus
D. Precentral
E. Postcentral
F. Insular

Item Description: SOM.MK.I.BPM2.4.NB.4.NSCI.0131 Postcentral gyrus homunculus

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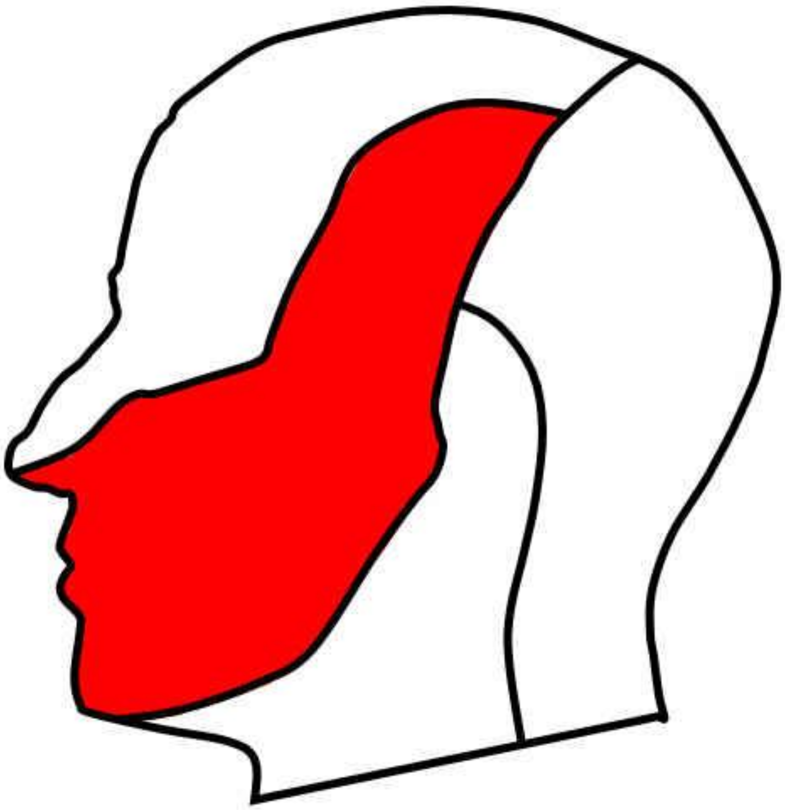
Question #: 8

A 27-year-old man comes to the physician because of a 6-month history of short, unilateral attacks of severe, shooting pain, that feel "like an electric shock" in the area marked in red in the schematic drawing attached. Each pain attack lasts only a few seconds, but they can cluster, followed by pain-free intervals lasting for several weeks. The pain attacks can be triggered by light touch in certain areas and by eating or talking. MRI of the head is normal. Dental conditions are normal. The pain is refractory to pharmacological intervention. Surgical intervention is considered, most likely involving the lesioning of pseudounipolar cell bodies lying lateral to the pons. Which of the following cranial nerves is most likely affected?

A. II
B. III
C. IV
D. V
E. IX

Attachment:



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Item Description: SOM.MK.IV.BPM2.4.NB.4.NSCI.0126 Pain management

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Question #: 9

A 37-year-old woman is brought to the emergency department because of sudden-onset of severe pain in the left chest and down the left arm. She is a heavy smoker and has high blood pressure. Her pain is suspected to be of myocardial ischemic origin. Which of the following structures most likely houses the cell bodies of the primary afferent fibers carrying the pain signals?

A. Left dorsal root ganglion
B. Right dorsal root ganglion
C. Left dorsal horn of the spinal cord
D. Right dorsal horn of the spinal cord
E. Left ventral posterolateral thalamic nucleus
F. Right ventral posterolateral thalamic nucleus
G. Left ventral posteromedial thalamic nucleus
H. Right ventral posteromedial thalamic nucleus

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0120 Pain types

Question #: 10

A 35-year-old-man comes to the physician because of a 4-day history of pain and itchiness affecting his chest. Physical examination shows an area of blistering on the chest (see photo). The patient reports a constant burning pain sensation originating from the blistered area. Which spinal segment is most likely affected?

A. Left C5
B. Left S5
C. Left T3
D. Right C5
E. Right S5
F. Right T3

Attachment:

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Item Description: SOM.MK.I.BPM2.4.NB.4.NSCI.0119 Somatosensation

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Question #: 11

A 6-year-old boy is brought to the physician by his father because of a 30-minute history of a cut on his foot. In preparation for stitching of the wound, lidocaine is administered by local injection. Insertion of the needle into his skin evokes a sudden jab of pain. Which of the following tracts/systems most likely mediates the painful sensation?

A. Corticospinal tract
B. Corticorubral tract
C. Anterolateral system
D. Medial lemniscus
E. Dorsal columnar

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0121 Pain pathway

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Question #: 12

A 27-year-old-man comes to the physician because of a 2-month history of tingling of the hands. He says he was involved in a car crash 6 months ago and hurt his neck. MRI shows a spinal abnormality (see image). Neurologic testing most likely shows which of the following patterns of sensory findings?

A. Left loss of pain and temperature; touch normal
B. Left loss of touch; pain and temperature normal
C. Left loss of temperature; pain and touch normal
D. Right loss of pain and temperature; touch normal
E. Right loss of touch; pain and temperature normal
F. Right loss of temperature; pain and touch normal
G. Bilateral loss of pain and temperature; touch normal
H. Bilateral loss of touch; pain and temperature normal
I. Bilateral loss of temperature; pain and touch normal

Attachment:

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Item Description: SOM.MK.I.BPM2.4.NB.4.NSCI.0119 Somatosensation

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Question #: 13

A 23-year-old-woman is brought to the emergency room because of a 1-hour history of traumatic injuries stemming from a car crash. Neurologic examination shows the absence of pain sensations in the left leg and the absence of touch sensations in the right leg. Sensation from the arms is unaffected. Which of the following structures is most likely damaged?

A. Left thoracic dorsal horn
B. Left cervical dorsal horn
C. Right thoracic dorsal horn
D. Right cervical dorsal horn
E. Left thoracic spinal hemisphere
F. Right thoracic spinal hemisphere
G. Left cervical spinal hemisphere
H. Right cervical spinal hemisphere

Item Description: SOM.MK.I.BPM2.4.NB.4.NSCI.0119 Somatosensory system

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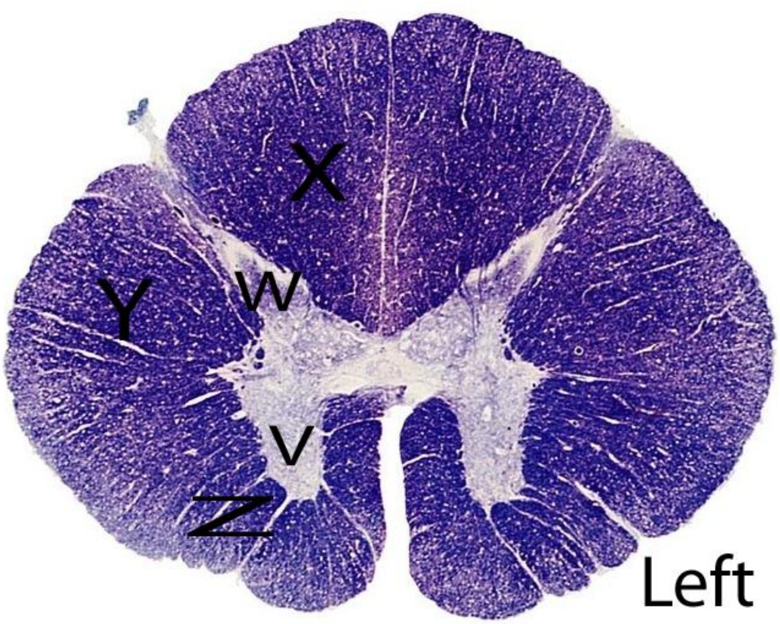
Question #: 14

A 42-year old woman is brought to the physician by her brother because of a 3-day history of stumbling in the dark. Neurological examination shows unsteadiness when she stands with her feet together and her eyes closed. She is insensitive to vibrating stimuli applied to her feet. Which of the structures indicated in the attached image is most likely injured bilaterally?

A. V
B. W
C. X
D. Y
E. Z

Attachment:



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Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0117 Somatosensory pathways

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Question #: 15

A 21-year-old woman is brought to the emergency department because of a 1-hour history of tingling of the left hand. MRI shows a lesion in the postcentral gyrus. Which of the following somatosensory modalities is/are most likely disrupted in this patient?

A. Touch
B. Pain
C. Thermoception
D. Vibration
E. Touch and pain

Item Description: SOM.CS.I.BPM2.4.NB.4.NSCI.0133 Somatosensory testing

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Question #: 16

An 11-year-old boy is brought to the physician because of a 3-year history of muscle weakness. He is unable to climb stairs and becomes fatigued easily. One of his maternal uncles had similar complaints and died at a young age. Physical examination shows difficulty in getting up from a supine position. Neurological examination shows hyporeflexia and decreased muscle strength. Genetic studies show mutations in the dystrophin gene. Which of the following structures is most likely lesioned?

A. Extrafusal fibers
B. Muscle spindle
C. Alpha motor neuron
D. II fibers
E. Ib fibers
F. Ia fibers

Item Description: SOM.MK.IV.BPM2.4.NB.2.NSCI.0164 Areflexia and hyporeflexia

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Question #: 17

A 29-year-old man comes to the physician because of a 3-week history of problems swallowing and speaking. Neurological examination shows atrophy and hypotonia of the left side of the tongue with fasciculations. Which of the following conditions is most likely?

A. Guillain-Barre disease
B. Myasthenia gravis
C. Lambert-Eaton syndrome
D. Duchenne muscular dystrophy
E. Amyotropic lateral sclerosis

Item Description: SOM.MK.II.BPM2.4.NB.2.NSCI.0173 Motor unit lesions of cell bodies or axons

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Question #: 18

A 52-year-old man comes to the physician because of a 2-week history of weakness. Neurological examination shows waxing strength of the grip bilaterally. He is treated successfully with 4-aminopyridine. His weakness is most likely linked to which of the following pathological processes?

A. Trauma
B. Gastrointestinal infection
C. Exposure to lead
D. Tetanus toxemia
E. Pulmonary tumor

Item Description: SOM.MK.II.BPM2.4.NB.2.NSCI.0174 Demyelination in the motor unit

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Question #: 19

A 67-year-old man comes to the physician because of fatigue and blurry vision that worsen in the afternoon. He says that he recovers after resting. Which of the following findings is most likely?

A. Waning in EMG
B. Fasciculations
C. Fibrillations
D. Hypertonia
E. Muscle wasting

Item Description: SOM.MK.II.BPM2.4.NB.2.NSCI.0178 Manifestations of motor unit disease

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Question #: 20

A 7-year-old boy is brought to the physician by his father because of problems rising to a standing position after lying down. When lying on his back on the floor, he first rolls onto his belly, then struggles to rise, eventually using his hands to push against his knees. His calves are hypertrophic. Which of the following diagnoses is most likely?

A. Dystonia
B. Slow-channel syndrome
C. Myotonia congenita
D. Upper motor neuron syndrome
E. Duchenne muscular dystrophy

Item Description: SOM.MK.II.BPM2.4.NB.2.NSCI.0176 Somatic Muscle lesions

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Question #: 21

A 65-year-old man is brought to the physician by his son because of a 1-week history of right hemiparesis. He has a history of hypertension and diabetes. Neurologic examination shows weakness and hyperreflexia of the right upper and lower limbs. Which of the following structures is most likely damaged?

A. Left internal capsule
B. Left posterior paracentral lobule
C. Right internal capsule
D. Right anterior paracentral lobule
E. Right postcentral gyrus
F. Left lumbar spinal cord

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0149 Lateral corticospinal pathway

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Question #: 22

A 56-year-old man is brought to the physician by his spouse because of a 4-hour history of uncontrolled rotational movements of his right leg. A diagnosis of hemiballismus is made. Which of the following sites is most likely injured?

A. Left frontal lobe
B. Left striatum

C. Left subthalamus
D. Left cerebellum
E. Left tectum
F. Right frontal lobe
G. Right striatum
H. Right subthalamus
I. Right cerebellum
J. Right tectum

Item Description: SOM.MK.II.BPM2.4.NB.2.NSCI.0191 Hyperkinesia direct and indirect

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Question #: 23

A 48-year-old man comes to the physician because of a 6-hour history of clumsiness. Neurologic examination shows dysarthria, left-sided dysdiadochokinesia, intention tremor, dysmetria and ataxia while attempting to perform skilled movements. Strength and sensory systems are normal. Which of the following cerebellar zones is most likely injured?

A. Left vestibulocerebellum
B. Left spinocerebellum
C. Left cerebrocerebellum
D. Right vestibulocerebellum
E. Right spinocerebellum
F. Right cerebrocerebellum

Item Description: SOM.MK.I.BPM2.4.NB.2.NSCI.0202 Cerebellar lesions and manifestations

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Question #: 24

A 67-year-old man is brought to the physician because of a 16-year history of Parkinson disease. Which of the following manifestations is/are most likely in this patient?

A. Dance-like movements
B. Falls
C. Titubation
D. Athetosis
E. Intention tremor

Item Description: SOM.MK.II.BPM2.4.NB.2.NSCI.0190 Parkinson direct and indirect

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Question #: 25

A 56-year-old woman is brought to the emergency room because she is unable to move her right leg. She is non-compliant with her anti-hypertensive medication. Physical examination shows a pulse rate of 100 bpm and blood pressure of 160/95 mmHg. She is conscious and oriented. Neurological examination shows hyperreflexia at the ankle and knee. Sensory testing shows losses of pain, temperature and vibratory sensations in the same leg. Cranial nerve testing is unremarkable. Which of the following additional neurological findings is most likely?

A. Flexor plantar response
B. Fasciculation
C. Fibrillations
D. Hypertonia
E. Tremors

Item Description: SOM.MK.II.BPM2.4.NB.2.NSCI.0152 Lower motor neuron lesions

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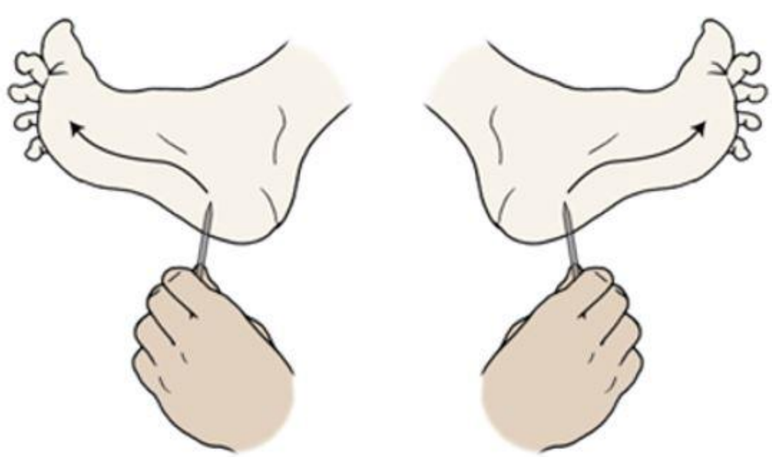
Question #: 26

A 6-month-old boy undergoes a routine developmental checkup. The bilateral plantar response is illustrated in the schematic drawing attached. Based on these findings, which of the following bilateral conditions is most likely in this boy?

A. Cortical lesion
B. Internal capsule lesion
C. Brainstem lesion
D. Spinal cord lesion
E. Peripheral nerve lesion
F. Normal development

Attachment:



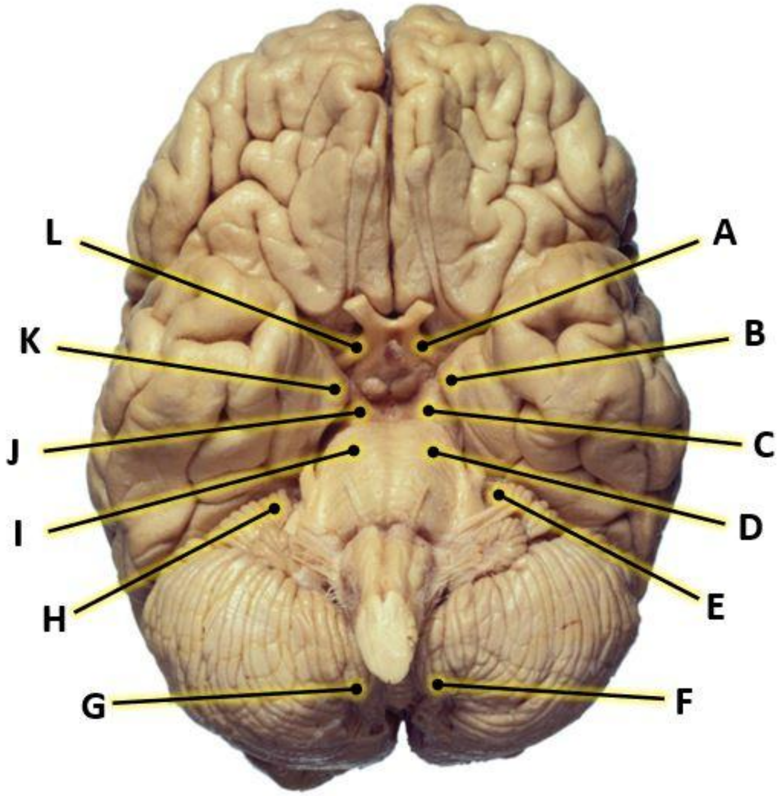
Item Description: Plantar response during 6 month of age

Question #: 27

A 44-year-old man is brought to the emergency department because of weakness on the left side of his body, including upper and lower extremities that had developed over the past few months. Neurologic examination confirms a left hemiplegia. The pupil of his right eye is abnormally dilated and responds sluggishly to light. Right eye movements are deficient. Herniation of which of the structures marked in the brain image attached through the tentorial notch is the most likely cause of the pathological findings in this patient?

A. A
B. B
C. C
D. D
E. E
F. F
G. G
H. H
I. I
J. J
K. K
L. L

Attachment:



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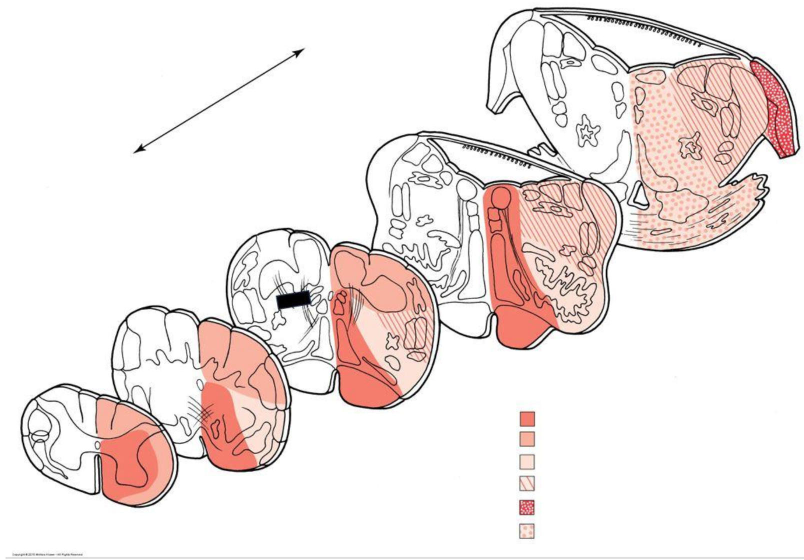
Item Description: Location of the uncus, uncal herniation

Question #: 28

A 33-year-old man comes to the physician because of a sudden onset of sensory deficits. Imaging shows a lesion indicated by a black bar in the schematic brainstem sections attached. Which of the following sensory deficits are most likely in this patient?

A. Ipsilateral loss of pain and temperature sensations of the body
B. Contralateral loss of pain and temperature sensations of the body
C. Ipsilateral loss of pain and temperature sensations of the face
D. Contralateral loss of pain and temperature sensations of the face
E. Ipsilateral loss of touch, vibration and proprioception sensations of the body
F. Contralateral loss of touch, vibration and proprioception sensations of the body
G. Ipsilateral loss of touch, vibration and proprioception sensations of the face
H. Contralateral loss of touch, vibration and proprioception sensations of the face

Attachment:



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Item Description: Internal arcuate fiber lesion

Question #: 29

A 23-year-old man in the intensive care unit has suddenly developed rigidity with bilateral extension of lower extremities, trunk and neck musculature. Upper extremities are flexed bilaterally at the elbow and wrist. Which of the following tracts are most likely intact bilaterally and responsible for the upper limb findings in this patient?

A. Corticorubral
B. Corticoreticular
C. Rubrospinal
D. Reticulospinal
E. Vestibulospinal

Item Description: Decorticate rigidity

Question #: 30

A 21 year old man come to the the physician for a general physical examination for insurance purpose. Physicial examination is with in normal limits. What is most likely seen on normal tympanic membrane examination?

A. Visible base of the light refled towards the anterior aspect
B. Head of the malleus visible
C. Stapes footplate visible
D. Malleoincudal joint visible
E. Chorda tympani nerve visible

Item Description: Tympanic membran